ವಿಶ್ವೇಶ್ವರಯ್ಯ ತಾಂತ್ರಿಕ ವಿಶ್ವವಿದ್ಯಾಲಯ

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

(A State University of Government of Karnataka Established as per the VTU Act, 1994)
"Jnana Sangama", Belagavi - 590018, Karnataka, India

QUC Quantum Club VTU Belagavi

INSTITUTION'S INNOVATION COUNCIL (Ministry of Education Initiative)

Visvesvaraya Research & Innovation Foundation

PRESENTS

QUANTUM STUDENTS' SUMMIT 2026

A NATIONAL LEVEL EVENT

THEME :
SYNERGIZING HARDWARE DEVELOPMENT FOR REALIZING QUANTUM TECHNOLOGY

16TH-17TH APRIL, 2026

DR. A. P. J. ABDUL KALAM AUDITORIUM, JNANA SANGAMA,
VTU MAIN CAMPUS, **BELAGAVI-18**, KARNATAKA

About VTU

Visvesvaraya Technological University (VTU), Belagavi, was established by the Government of Karnataka in 1998 under the VTU Act, 1994 to strengthen and promote technical education in the state. The headquarters is located in Belagavi, with more than 220 affiliated engineering and technological institutions, VTU plays a pivotal role in shaping quality engineers and technologists. The university actively promotes skill-based and industry-oriented education, while also establishing several Centres of Excellence (CoEs) in emerging areas of science and technology to foster advanced research, innovation, and industry collaboration..

About VTU Quantum Club

The Quantum Club at Visvesvaraya Technological University is a vibrant student driven community dedicated to fostering innovation, collaboration, and leadership in emerging technologies. The club serves as a platform for students to explore cutting-edge fields such as quantum computing, artificial intelligence, and advanced engineering sciences, while also encouraging interdisciplinary learning and creative problem-solving. Through workshops, seminars, and national-level events, the Quantum Club brings together bright minds from across India to share knowledge, showcase talent, and build networks that extend beyond the classroom. With its focus on empowering students to think critically and lead with vision, the Quantum Club has become a hub of energy, ideas, and inspiration for the next generation of technologists and innovators.

About the Quantum Student Summit

The Quantum Club of Visvesvaraya Technological University (VTU), Belagavi is organizing the Quantum Student Summit 2026, an exclusive national event dedicated to advancing knowledge in quantum Science and technology. It provides a unique opportunity for students to present ideas, exchange knowledge, and engage with cutting-edge quantum research.

Key Objectives

The National Level Quantum Student Summit at VTU aims to empower students and researchers to lead India's quantum future by fostering innovation, collaboration, and skill development in alignment with the National Quantum Mission.

- Empower students as pioneers in quantum research and entrepreneurship.
- Synergize quantum hardware development through Interactive Learning
- Facilitate collaboration between academia, industry, and innovators.
- Align student initiatives with India's National Quantum Mission (2023–2031).
- Build capacity by training faculty and students across VTU's network.
- Create a sustainable ecosystem for quantum education and workforce readiness.

Events

POSTER PRESENTATION | QUBIT-THON

PLENARY TALKS | PANEL DISCUSSION

Poster Presentation Tracks

Track 1: Quantum Hardware & Architectures
Scalable Quantum Processor Architectures (ion trap, superconducting, photonic)
Quantum circuits and its application
Charge Detection and Readout Techniques
Cryogenics electronics

Track 2: Network security
Quantum Cryptography for Post-Quantum Cybersecurity
Quantum Key Distribution (QKD) for National Security Networks
Quantum-Safe Blockchain Systems
Secure IoT Using Quantum Encryption

Track 3: Space
Entanglement-Based Space Communication
Quantum Sensors for Submarine Detection
Quantum LIDAR for High-Resolution Surveillance
Quantum-Secure Battlefield Communication

Track 4: Quantum AI and Data Science
Hybrid Quantum–Classical Machine Learning Models
Quantum Support Vector Machines (QSVM)
Quantum Reinforcement Learning
Quantum Kernel Methods
Quantum Machine Learning Algorithms for Big Data
Quantum Neural Networks (QNNs)

Track 5: Quantum Physics
Quantum Measurement Problem and Decoherence
Quantum Tunneling in Nano-Scale Devices
Quantum Thermodynamics
Quantum Transport in Superconductors
Superconductivity in Low-Dimensional Systems
Quantum Phase Estimation (QPE)

Track 6: Quantum Chemistry and Materials
Quantum materials and devices:
2D Quantum Materials
Spin–Orbit Coupling and Spintronic Materials
Quantum Hall and Fractional Quantum Hall Materials
Harrow-Hassidim-Lloyd algorithm (HHL)
Variational Quantum Eigensolver (VQE)

QUBIT-THON (IDEATHON) Problem Statements

Domain-Electricals

1. How can quantum Technology techniques be used to optimize battery materials, charging strategies, and energy management to create longer-lasting and more efficient energy storage systems.

Domain- Banking

2. How can quantum algorithms be used to improve the accuracy and efficiency of fraud detection systems in financial and digital transactions compared to classical machine learning approaches.

Domain-Healthcare

3. How can Quantum Graph Neural Networks be designed and applied to effectively classify healthcare data while capturing complex relationships in medical datasets and improving efficiency compared to classical approaches.

Domain-Smart Industry

4. How can quantum optimization techniques be applied to improve scheduling efficiency in smart factories while reducing delays, energy consumption, and operational costs.